Below is the detailed Software Requirements Specification (SRS) and Functional Architecture Blueprint for the **Unified Insurance Platform**, designed to support Life, General (Non-Life), and Health insurance within a single, cloud-ready, strictly tenant-isolated platform.

UNIFIED INSURANCE PLATFORM - INSYS CORE

Software Requirements Specification (SRS) and Functional Architecture Blueprint

Version: 1.0
Date: 21 October 2026
Status: Proposed baseline for stakeholder review and engineering elaboration
Product scope: Life Insurance, General Insurance, and Health Insurance
Architecture: Multi-tenant, cloud-ready, configuration-driven, domain-oriented

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PART I — SOFTWARE REQUIREMENTS SPECIFICATION

1. Executive Summary

1.1 Purpose

The Unified Insurance Platform (UIP) is an enterprise-grade insurance administration platform intended to support multiple insurers, insurance intermediaries, and insurance businesses through a common technology platform.

The platform will support three primary insurance domains:

1. **Life Insurance**
2. **General Insurance (Non-Life)**
3. **Health Insurance**

The platform will share common capabilities such as customer management, product configuration, pricing frameworks, workflow, billing, payments, financial accounting, document management, integration, reporting, and security.

However, each insurance domain will retain ownership of its policies, claims, business rules, and transaction lifecycles.

The objective is to achieve platform-level reuse without forcing fundamentally different insurance processes into one generic business model.

1.2 Core architectural decisions

The following decisions are established as baseline requirements.

Area	Decision	Implication
Multi-tenancy	Strict tenant isolation	No tenant may access another tenant's data or unauthorised resources.
Cloud deployment	Required	The platform must support automated cloud deployment, scaling, monitoring, backup, and disaster recovery.
Policy model	Separate domain-owned policy aggregates	Life, General, and Health will have distinct policy models and lifecycles.
Product configuration	Primarily configuration-driven	Routine product variations must not require software development.
Shared services	Common platform capabilities	Domains reuse shared infrastructure and services where their semantics genuinely align.

Integration	API-first and event-enabled	The platform supports external integrations and asynchronous processing.
Security	Security by design	Identity, authorisation, tenant isolation, audit, and encryption are foundational.
Product governance	Versioned and controlled	Product changes require validation, approval, and effective-date management.

1.3 Business objectives

The platform shall:

- Reduce the time and cost required to launch insurance products.
- Allow multiple insurers to operate securely on a common platform.
- Support multiple products, countries, currencies, and distribution channels.
- Reduce duplicated functionality across insurance lines.
- Maintain independent Life, General, and Health business processes.
- Support the full policy lifecycle, from quotation through servicing and termination.
- Support claims processing, adjudication, settlement, and financial reconciliation.
- Provide auditable and reproducible pricing and underwriting decisions.
- Integrate with banks, payment providers, brokers, agents, healthcare providers, reinsurers, and accounting systems.
- Provide a foundation for future insurance products without requiring a complete platform redesign.

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2. Scope of the Platform

2.1 Functional scope

The platform will include the following major capability groups.

Capability group	Scope
Platform foundation	Tenant management, identity, access control, audit, configuration governance
Party and customer management	Individuals, organisations, customers, insured parties, beneficiaries, intermediaries, providers
Product Factory	Product definitions, benefits, coverages, rates, eligibility, underwriting rules, workflows
Life Insurance	Proposals, underwriting, policies, benefits, premiums, maturity, surrender, life claims
General Insurance	Quotation, risk objects, underwriting, policies, endorsements, renewals, claims
Health Insurance	Schemes, members, dependants, benefits, providers, preauthorisation, medical claims
Shared claims framework	Claim registration, case management, documents, workflow, audit, settlement integration

Billing and receivables	Premium schedules, invoices, collections, allocations, arrears
Payments and settlements	Payment initiation, status tracking, refunds, reconciliation, claim payments
Distribution	Brokers, agents, bancassurance, digital channels, partner distribution
Financial management	Commissions, financial subledger, reconciliation, general ledger integration
Reinsurance	Treaties, facultative arrangements, cessions, recoveries
Document management	Policy schedules, certificates, endorsements, claim evidence, correspondence
Integration	APIs, webhooks, events, external system adapters
Compliance and reporting	Audit, regulatory extracts, operational reporting, data governance
Analytics	Portfolio, underwriting, claims, financial and operational analytics

2.2 Initial exclusions

Unless separately approved, the initial platform will not attempt to replace:

- An insurer’s complete enterprise general ledger.
- External banking and payment networks.

- A hospital's clinical information system.
- External actuarial modelling systems where specialised calculations are required.
- Every jurisdiction-specific regulatory reporting package from day one.
- Every possible insurance product through configuration alone.

The architecture must nevertheless provide extension points for these capabilities.

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3. Stakeholders and User Roles

The platform must support multiple user groups, with permissions restricted by tenant, organisation, role, and business responsibility.

User role	Principal responsibilities
Platform Administrator	Platform configuration, tenant provisioning, platform operations
Tenant Administrator	Insurer-specific users, roles, settings, organisation structures
Product Manager	Product configuration, product versioning, publication
Product Actuary	Rating assumptions, premium parameters, actuarial validation
Underwriter	Risk assessment, referrals, acceptance, loading and rejection
Policy Administrator	Policy issuance, servicing, endorsements, renewal and cancellation
Claims Assessor	Claims assessment, reserves, adjudication, approval

Claims Manager	Delegated authority, escalation, exception approval
Finance Officer	Billing, collections, commission, reconciliation, financial posting
Broker or Agent	Quotation, proposal submission, permitted policy servicing
Healthcare Provider	Eligibility verification, preauthorisation and claim submission
Customer or Member	Access to permitted policy, benefit, payment and claim information
Compliance Officer	Regulatory reporting, audit review and compliance monitoring
Internal or External Auditor	Controlled access to audit evidence and transaction histories
Integration Service	Approved machine-to-machine transactions

Requirement SRS-001: Role-based access

The platform MUST support configurable role-based access control.

Requirement SRS-002: Segregation of duties

The platform MUST support configurable maker-checker controls for sensitive operations, including:

- Product publication.
- Underwriting overrides.
- Claim approval.

- Payment release.
- Financial adjustments.
- Privileged user administration.

Requirement SRS-003: Tenant-scoped permissions

Permissions MUST be restricted to authorised tenants and, where configured, to specific products, branches, organisations, and transaction categories.

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4. Multi-Tenancy and Tenant Isolation

Strict tenant isolation is a foundational architectural requirement, not an optional product feature.

4.1 Tenant Management module

Submodules

1. Tenant registration and onboarding.
2. Tenant profile management.
3. Tenant configuration.
4. Tenant-specific product availability.
5. Tenant user and role administration.
6. Tenant organisation structures.
7. Tenant-specific integration configuration.
8. Tenant feature entitlements.
9. Tenant operational limits.
10. Tenant suspension and reactivation.
11. Tenant offboarding and termination.
12. Tenant audit and operational monitoring.

Functional requirements

TEN-001: Every tenant MUST have a unique, immutable identifier.

TEN-002: Tenant context MUST be derived from a trusted authentication or service identity mechanism. An untrusted request parameter alone MUST NOT determine the tenant.

TEN-003: Every tenant-scoped operation MUST validate that the authenticated principal is authorised for the resolved tenant.

TEN-004: The platform MUST enforce tenant isolation across transactional data, documents, APIs, caches, events, search, analytics, reporting, and background processing.

TEN-005: Cross-tenant access attempts MUST be rejected and logged.

TEN-006: Tenant-specific product configurations, user permissions, integration credentials, and operational settings MUST remain isolated.

TEN-007: Administrative access to tenant data MUST be controlled, auditable, and restricted to an approved operational purpose.

TEN-008: Tenant onboarding, suspension, reactivation, and termination MUST be governed by defined workflows.

4.2 Tenant isolation architecture

Isolation must be implemented through multiple independent controls.

Layer	Required control
Authentication	Resolve tenant context from trusted identity information
Authorisation	Verify tenant and resource-level permissions
Application	Enforce tenant scope within domain services
Database	Use tenant-aware access controls and an approved isolation strategy
Documents	Enforce tenant-specific access to object storage and document metadata
Messaging	Validate tenant identity in event producers and consumers

Cache	Use tenant-aware cache keys and prevent cross-tenant data reuse
Search	Restrict queries and indexed documents by tenant
Reporting	Apply tenant-aware data access and export restrictions
Audit	Record tenant identity for relevant operations
Operations	Control tenant-specific diagnostics, backup, restoration, and support access

Important: Application-level tenant filtering alone is insufficient. The architecture should use database controls and other independent safeguards to reduce the consequences of application defects.

4.3 Supported deployment models

The platform should support three tenant isolation profiles.

Profile A: Shared infrastructure

Multiple tenants share application infrastructure and database services, with strict logical separation.

Suitable for tenants whose security, regulatory, and contractual requirements permit this model.

Profile B: Dedicated tenant database

A tenant receives a dedicated database or equivalent isolated data service while sharing selected application services.

Suitable for customers requiring stronger data separation.

Profile C: Dedicated tenant environment

A tenant receives a dedicated application and data environment.

Suitable for customers requiring stronger infrastructure separation, specific hosting arrangements, or specialised operational controls.

These profiles should be selectable according to risk, regulatory requirements, data residency, performance, and commercial service tier.

Strict tenant isolation does not automatically require separate physical infrastructure for every tenant. It requires demonstrable enforcement of tenant boundaries, with stronger isolation options available where justified.

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5. Identity, Access, and Security

5.1 Identity and Access Management module

Submodules

- User registration and administration.
- Authentication.
- Single sign-on.
- Multi-factor authentication.
- Role-based access control.
- Attribute-based access policies where needed.
- Service-account management.
- API client credentials.
- Session management.
- Credential rotation.
- Privileged access management.
- Access review and revocation.
- Authentication and security audit.

Requirements

IAM-001: The platform MUST support tenant-scoped user identities and permissions.

IAM-002: It MUST support federation with approved identity providers using standard protocols such as OpenID Connect or SAML where required.

IAM-003: Multi-factor authentication MUST be supported for privileged and sensitive roles.

IAM-004: Service accounts MUST have explicit scopes, limited permissions, and controlled credentials.

IAM-005: The system MUST support prompt revocation of compromised or deactivated identities.

IAM-006: Sensitive operations MUST record the acting user or service identity, tenant, operation, target resource, timestamp, and outcome.

IAM-007: Authorisation MUST be enforced by backend services and MUST NOT rely exclusively on user-interface restrictions.

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6. Shared Party and Customer Management

The Party Management module provides common party identities and relationships used across all insurance domains.

It must not become a universal data model that absorbs every domain-specific attribute.

6.1 Submodules

1. Individual party management.
2. Organisation management.
3. Customer profiles.
4. Contact information.
5. Address management.
6. Party relationships.
7. Identity documents and verification references.
8. Consent and communication preferences.
9. Duplicate detection.
10. Party search.
11. Customer status and history.
12. Party-role management.

6.2 Party roles

A party may have multiple roles, including:

- Policyholder.
- Insured person.
- Beneficiary.

- Claimant.
- Payer.
- Payee.
- Broker.
- Agent.
- Employer.
- Healthcare provider.
- Reinsurer.
- Corporate customer.

The same individual could, for example, be a policyholder on a Life policy and a dependant on a Health policy.

The platform should maintain a common party identity while keeping domain-specific relationships and permissions within the appropriate domain.

Requirements

PTY-001: The system MUST support individuals and organisations.

PTY-002: A party MUST be capable of holding multiple roles.

PTY-003: The system MUST support tenant-scoped party access and controlled sharing of information.

PTY-004: Duplicate detection and record merging SHOULD be supported with an auditable history.

PTY-005: Personal information MUST be collected and exposed according to authorised business purposes and applicable privacy requirements.

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7. Product Factory and Configuration Management

This is one of the platform's most strategically important components.

The Product Factory enables insurers to configure products, benefits, coverage, pricing, eligibility, underwriting rules, and workflows without requiring code changes for routine variations.

7.1 Product Catalogue

Submodules

- Product registration.
- Product classification.
- Product family management.
- Product versioning.
- Product lifecycle management.
- Product availability by tenant.
- Market and country applicability.
- Distribution-channel applicability.
- Product approval.
- Product publication.
- Product withdrawal and retirement.

7.2 Coverage and benefit configuration

Submodules

- Coverage definitions.
- Benefit definitions.
- Coverage limits.
- Benefit limits.
- Sublimits.
- Exclusions.
- Deductibles and excesses.
- Waiting periods.
- Optional benefits.
- Benefit combinations.
- Benefit dependencies.
- Eligibility criteria.
- Coverage effective dates.

7.3 Pricing and rating configuration

Submodules

- Rating factors.
- Rating tables.
- Premium parameters.
- Risk classifications.

- Discounts.
- Loadings.
- Minimum and maximum premiums.
- Taxes and levies.
- Commission parameters.
- Effective-dated pricing.
- Rating simulation.
- Rating audit.

7.4 Underwriting configuration

Submodules

- Eligibility rules.
- Underwriting questionnaires.
- Risk thresholds.
- Referral rules.
- Automatic acceptance rules.
- Medical evidence requirements.
- Document requirements.
- Approval thresholds.
- Decline conditions.
- Underwriting authority matrices.

7.5 Workflow configuration

The platform should allow authorised users to configure workflows for:

- Proposal approval.
- Underwriting referral.
- Policy issuance.
- Policy endorsements.
- Renewal.
- Claims assessment.
- Claims approval.
- Payment release.
- Exception handling.
- Escalation.

7.6 Product document configuration

The platform should support versioned templates for:

- Policy wording.
- Policy schedules.
- Certificates.
- Endorsements.
- Premium notices.
- Renewal notices.
- Claims correspondence.
- Benefit schedules.
- Health membership documents.

7.7 Configuration lifecycle

A product configuration should progress through controlled states:

Draft → Validated → Submitted for Approval → Approved → Published → Effective → Withdrawn or Retired

The exact workflow may vary by tenant.

Requirements

PRD-001: Product definitions MUST be versioned.

PRD-002: Published product versions MUST be immutable.

PRD-003: Changes to published products MUST create a new version.

PRD-004: The system MUST validate product configuration before publication.

PRD-005: The platform MUST support simulation and test cases before publication.

PRD-006: Publication MUST support maker-checker approval.

PRD-007: Product versions MUST support effective dates.

PRD-008: Transactions MUST retain references to the product, rating, and rules versions used.

PRD-009: The system MUST identify missing parameters, invalid values, incompatible benefits, and unresolved dependencies.

PRD-010: Configuration changes MUST be auditable.

7.8 Configuration versus software development

The platform should distinguish routine product configuration from genuinely new domain functionality.

Change	Expected approach
Change premium rates	Configuration
Add a coverage limit	Configuration
Add a product variant	Configuration
Change eligibility thresholds	Configuration
Add an exclusion	Configuration
Change underwriting referral rules	Configuration
Change policy wording	Template configuration
Introduce a new standard benefit type	Configuration if the domain model supports it
Introduce a fundamentally new actuarial calculation	New calculation component may be required
Introduce a new policy lifecycle	Domain extension may be required

Configuration must be typed, validated, versioned, and governed. Arbitrary scripts should not be the default mechanism for business variation.

PART II — FUNCTIONAL ARCHITECTURE BLUEPRINT

8. Logical Architecture

The platform should be organised into five layers.

Layer	Components	Responsibility
Experience	Insurer portal, broker portal, customer portal, provider portal, partner channels	User and partner interaction
API and Integration	API gateway, partner adapters, webhooks, event messaging	Secure system interaction
Shared Insurance Services	Party, Product Factory, Rating, Rules, Workflow, Documents, Billing, Payments	Reusable business capabilities
Insurance Domains	Life, General, Health	Domain-owned insurance operations
Enterprise and Platform Foundation	Subledger, Reinsurance, Compliance, Tenant Isolation, IAM, Observability, Infrastructure	Financial control and runtime services

8.1 Architectural principle

The platform should reuse common capabilities where their semantics align.

It should not force all insurance processes into generic models simply because they share terminology.

For example:

- All insurance domains may use the same document-generation framework.
- Life and General policies should not be forced into one identical policy aggregate.
- All domains may use the same workflow infrastructure.
- Life, General, and Health claims should have distinct adjudication engines.
- All domains may use a common rating execution framework.
- Domain-specific pricing calculations must remain distinct where their actuarial or underwriting logic differs.

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9. Life Insurance Domain

The Life Insurance domain owns its proposals, underwriting decisions, policies, benefits, lifecycle, and claims.

9.1 Life Insurance module hierarchy

A. Life Proposal Management

Submodules:

1. Proposal capture.
2. Applicant details.
3. Insured-life details.
4. Beneficiary nominations.
5. Product selection.
6. Coverage selection.
7. Premium quotation.
8. Disclosure questionnaires.
9. Supporting documents.
10. Proposal validation.
11. Proposal submission.
12. Proposal status tracking.

B. Life Underwriting

Submodules:

1. Underwriting rules.
2. Risk classification.
3. Medical evidence requirements.
4. Financial underwriting.
5. Occupation and lifestyle assessment.
6. Underwriting referrals.
7. Manual assessment.
8. Acceptance decisions.
9. Premium loadings.
10. Exclusions.
11. Postponement.
12. Decline management.

C. Life Policy Administration

Submodules:

1. Policy issuance.
2. Policy numbering.
3. Policyholder management.
4. Insured-life management.
5. Policy status management.
6. Policy endorsements and amendments.
7. Premium schedule management.
8. Policy document generation.
9. Policy servicing.
10. Policy history.
11. Lapse and reinstatement.
12. Surrender and termination.

D. Benefits Administration

Submodules:

1. Sum assured.
2. Death benefits.
3. Disability benefits.
4. Critical illness benefits.
5. Income protection benefits.
6. Maturity benefits.
7. Benefit eligibility.

8. Benefit limits.
9. Benefit exclusions.
10. Benefit calculation.

E. Premium Administration

Submodules:

1. Premium frequency.
2. Premium due schedules.
3. Premium adjustments.
4. Premium allocation interfaces.
5. Grace-period processing.
6. Arrears management.
7. Lapse processing.
8. Reinstatement processing.
9. Premium history.

F. Cash Values and Policy Loans

Where supported by the product:

1. Cash-value calculation.
2. Surrender-value calculation.
3. Policy-loan eligibility.
4. Loan issuance.
5. Loan balance management.
6. Interest calculation.
7. Repayment processing.
8. Partial withdrawals.
9. Financial reconciliation.

G. Beneficiary Management

Submodules:

1. Beneficiary registration.
2. Beneficiary identification.
3. Beneficiary allocation.
4. Beneficiary changes.
5. Beneficiary validation.
6. Beneficiary history.
7. Beneficiary verification during claims.

H. Life Claims

Submodules:

1. Claim notification.
2. Claim registration.
3. Claimant verification.
4. Beneficiary verification.
5. Policy and coverage validation.
6. Evidence collection.
7. Medical or death documentation.
8. Claims assessment.
9. Benefit calculation.
10. Approval and escalation.
11. Settlement.
12. Claim closure.

I. Maturity Management

Submodules:

1. Maturity identification.
2. Maturity notifications.
3. Eligibility verification.
4. Maturity calculation.
5. Payee validation.
6. Approval.
7. Payment instruction.
8. Settlement reconciliation.

J. Group Life

Submodules:

1. Group scheme administration.
2. Employer management.
3. Group membership.
4. Eligibility.
5. Member enrolment.
6. Scheme-level benefits.
7. Group premium calculations.
8. Member changes.
9. Group claims.

K. Life Actuarial Interfaces

Submodules:

1. Valuation data extraction.
2. Assumption version references.
3. Cash-flow data preparation.
4. Product and policy data exports.
5. Actuarial calculation integration.
6. Reconciliation.
7. Valuation reporting.

Life-domain requirements

LIF-001: The domain MUST support proposal capture and quotation.

LIF-002: Underwriting decisions MUST be traceable and reproducible to the extent permitted by retained data and calculation methods.

LIF-003: Issued policies MUST reference approved product and configuration versions.

LIF-004: Policy servicing MUST validate each requested change against policy state, product rules, and applicable authority.

LIF-005: Premium processing MUST support configured payment frequencies, grace periods, arrears, lapse, and reinstatement.

LIF-006: The domain MUST support product-appropriate death, disability, critical illness, maturity, and other benefit claims.

LIF-007: Cash-value, surrender-value, and loan capabilities MUST be enabled only for products that support them.

LIF-008: Life claims MUST validate policy status, insured-life details, benefit eligibility, exclusions, and claimant authority.

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10. General Insurance Domain

The General Insurance domain owns its risk objects, policies, endorsements, renewals, claims, and associated business rules.

10.1 General Insurance module hierarchy

A. Quotation and Proposal Management

Submodules:

1. Proposal capture.
2. Risk information.
3. Product selection.
4. Coverage selection.
5. Premium calculation.
6. Quote versioning.
7. Quote validity.
8. Document generation.
9. Proposal submission.
10. Quote acceptance.
11. Quote rejection or expiry.

B. General Underwriting

Submodules:

1. Risk assessment.
2. Eligibility rules.
3. Risk classification.
4. Underwriting questionnaires.
5. Referral rules.
6. Acceptance and rejection.
7. Deductible/excess selection.
8. Special conditions.
9. Approval authority.
10. Underwriting audit.

C. Risk Object Management

Submodules:

1. Insured-object registration.
2. Risk-object classification.
3. Risk attributes.
4. Risk location.
5. Risk valuation.
6. Risk-object coverage.
7. Risk-object history.
8. Risk aggregation.
9. Risk inspection references.
10. Risk-object changes.

Examples of risk objects include vehicles, properties, vessels, equipment, inventory, and liabilities.

D. General Policy Administration

Submodules:

1. Policy issuance.
2. Policy schedules.
3. Coverage attachment.
4. Premium schedule.
5. Endorsements.
6. Mid-term adjustments.
7. Renewals.
8. Cancellation.
9. Reinstatement where permitted.
10. Policy documents.
11. Policy history.
12. Policy servicing.

E. Coverage Management

Submodules:

1. Coverage definitions.
2. Coverage limits.
3. Deductibles and excesses.
4. Exclusions.
5. Extensions.
6. Conditions.
7. Coverage periods.
8. Sublimits.
9. Aggregation limits.
10. Coverage validation.

F. Motor Insurance

Submodules:

1. Vehicle information.
2. Registration and identification.
3. Vehicle valuation.
4. Driver information.
5. Motor risk assessment.
6. Comprehensive coverage.

7. Third-party liability.
8. Theft and accident coverage.
9. Fleet administration.
10. Motor claims.
11. Repair and assessment interfaces.
12. Vehicle-related endorsements.

G. Property Insurance

Submodules:

1. Property registration.
2. Property location.
3. Building valuation.
4. Contents valuation.
5. Property risk assessment.
6. Fire and allied perils.
7. Natural disaster coverage.
8. Business interruption.
9. Property endorsements.
10. Property claims.

H. Marine Insurance

Submodules:

1. Vessel or cargo risk.
2. Voyage information.
3. Transit information.
4. Cargo valuation.
5. Marine coverage.
6. Policy certificates.
7. Voyage endorsements.
8. Marine claims.
9. Salvage and recovery.

I. Commercial Insurance

Submodules:

1. Commercial customer.
2. Business risk profile.
3. Commercial package policies.
4. Liability insurance.
5. Equipment insurance.

6. Business interruption.
7. Grouped risk objects.
8. Commercial underwriting.
9. Commercial claims.
10. Renewal and servicing.

J. General Claims

Submodules:

1. First notification of loss.
2. Claim registration.
3. Policy and coverage validation.
4. Loss details.
5. Claimant and third-party management.
6. Evidence collection.
7. Loss assessment.
8. Surveyor or adjuster assignment.
9. Repair and replacement estimates.
10. Reserve management.
11. Deductible/excess calculation.
12. Settlement approval.
13. Payment instruction.
14. Salvage and recovery.
15. Claim closure.

K. General Reinsurance and Actuarial Interfaces

Submodules:

1. Risk exposure extraction.
2. Treaty allocation interfaces.
3. Facultative reinsurance references.
4. Claims recovery interfaces.
5. Actuarial data extraction.
6. Portfolio reporting.

General-domain requirements

GEN-001: The domain MUST support quotation, underwriting, and policy issuance.

GEN-002: Policies MUST support multiple insured risk objects where applicable.

GEN-003: Endorsements MUST support effective dates, coverage changes, premium adjustments, document generation, and audit history.

GEN-004: Renewals, cancellations, and reinstatements MUST follow configurable product rules.

GEN-005: Different product families MUST be able to use distinct risk attributes and rating logic.

GEN-006: Claims MUST support registration, coverage validation, assessment, reserves, approval, settlement, and closure.

GEN-007: The system SHOULD support loss adjusters, surveyors, repairers, salvage, and third-party recovery.

GEN-008: Claim settlement MUST account for deductibles, excesses, limits, and applicable valuation rules.

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11. Health Insurance Domain

The Health Insurance domain owns health plans, schemes, memberships, eligibility decisions, provider networks, preauthorisations, and medical claims.

Medical data requires additional access controls and careful data minimisation.

11.1 Health Insurance module hierarchy

A. Health Product and Plan Management

Submodules:

1. Health product catalogue.
2. Plan definitions.
3. Benefit schedules.
4. Coverage limits.
5. Sublimits.
6. Exclusions.
7. Waiting periods.
8. Co-payments.
9. Deductibles.
10. Provider-network restrictions.
11. Eligibility rules.
12. Product versioning.

B. Scheme and Group Administration

Submodules:

1. Employer and sponsor management.
2. Group scheme registration.
3. Scheme configuration.
4. Group membership.
5. Contribution schedules.
6. Member eligibility.
7. Scheme renewal.
8. Group billing.
9. Employer reporting.
10. Scheme-level administration.

C. Member Administration

Submodules:

1. Principal member registration.
2. Dependant registration.
3. Member enrolment.
4. Member identification.
5. Membership effective dates.
6. Membership changes.
7. Member suspension.
8. Reinstatement.
9. Member termination.
10. Membership history.
11. Membership-card generation.
12. Member portal integration.

D. Eligibility Management

Submodules:

1. Member eligibility verification.
2. Policy and scheme validation.
3. Benefit eligibility.
4. Coverage period validation.
5. Waiting-period validation.
6. Provider-network eligibility.
7. Available benefit limits.
8. Eligibility decision history.
9. Provider-facing eligibility APIs.

E. Benefit Management

Submodules:

1. Inpatient benefits.
2. Outpatient benefits.
3. Specialist benefits.
4. Pharmaceutical benefits.
5. Dental benefits.
6. Optical benefits.
7. Maternity benefits.
8. Chronic-care benefits.
9. Emergency benefits.
10. Annual and lifetime limits.
11. Co-payment rules.
12. Exclusion and waiting-period rules.

The exact benefit categories will depend on the insurer's product portfolio.

F. Provider Network Management

Submodules:

1. Provider registration.
2. Provider verification.
3. Provider classification.
4. Provider specialties.
5. Provider locations.
6. Provider contracts.
7. Network participation.
8. Effective-dated tariffs.
9. Provider suspension.
10. Provider performance.
11. Provider payment details.

G. Preauthorisation Management

Submodules:

1. Preauthorisation request.
2. Member eligibility validation.
3. Service and procedure details.
4. Supporting medical documentation.
5. Benefit validation.
6. Clinical review workflow.
7. Approval and rejection.
8. Conditional approval.

9. Authorised limits.
10. Authorisation expiry.
11. Modification and cancellation.
12. Preauthorisation history.

H. Medical Claims

Submodules:

1. Provider claim submission.
2. Member reimbursement claims.
3. Claim registration.
4. Member verification.
5. Provider verification.
6. Service and tariff validation.
7. Benefit-limit validation.
8. Preauthorisation matching.
9. Duplicate claim detection.
10. Medical coding integration.
11. Claim adjudication.
12. Co-payment calculation.
13. Fraud indicators.
14. Approval and escalation.
15. Payment and reconciliation.

I. Utilisation Management

Submodules:

1. Utilisation rules.
2. High-cost service review.
3. Referral management.
4. Repeat-service checks.
5. Benefit-limit monitoring.
6. Case management.
7. Clinical review workflow.
8. Exception handling.
9. Utilisation reporting.

J. Health Payments

Submodules:

1. Provider payment calculation.
2. Member reimbursement.

3. Payment instruction generation.
4. Payment approval.
5. Settlement processing.
6. Remittance advice.
7. Payment reconciliation.
8. Payment exception handling.

K. Health Fraud and Abuse Controls

Submodules:

1. Duplicate claim detection.
2. Unusual utilisation indicators.
3. Provider billing anomalies.
4. Repeated-service checks.
5. Suspicious claim patterns.
6. Investigation workflow.
7. Evidence management.
8. Case outcomes and audit.

Automated indicators should support authorised investigation and review. They should not automatically make high-impact clinical or claims decisions without appropriate governance.

Health-domain requirements

HLT-001: The domain MUST support health products, plans, benefits, exclusions, waiting periods, co-payments, and limits.

HLT-002: The domain MUST support principal members, dependants, group schemes, enrolment, and membership lifecycle changes.

HLT-003: The platform MUST expose authorised member-eligibility inquiries.

HLT-004: Provider contracts and tariffs MUST support effective dating.

HLT-005: The platform MUST support preauthorisation requests, decisions, limits, expiry, and audit.

HLT-006: The domain MUST support provider-submitted claims and member reimbursement claims.

HLT-007: Medical claims MUST validate membership, coverage, benefit limits, applicable tariffs, preauthorisation, and configured exclusions.

HLT-008: Medical information MUST be restricted according to role, purpose, and minimum necessary access.

HLT-009: Provider and member payments MUST be traceable and reconcilable.

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12. Shared Claims Framework

The platform should provide common claims infrastructure without forcing all claims into identical business processes.

12.1 Shared claims capabilities

- Claim registration framework.
- Claim numbering.
- Claimant and beneficiary references.
- Document and evidence management.
- Case assignment.
- Work queues.
- Workflow and escalation.
- Audit history.
- Communications.
- Fraud indicators.
- Financial transaction interfaces.
- Payment instruction framework.
- Claims reporting.

12.2 Domain-specific claims engines

Life	General	Health
Death claims	Motor claims	Provider claims
Disability claims	Property claims	Reimbursement claims
Critical illness claims	Marine claims	Preauthorisation-linked claims

Maturity claims	Liability claims	Benefit-limit adjudication
Beneficiary verification	Loss adjustment	Tariff validation
Sum assured validation	Deductible/excess application	Co-payment calculation
Benefit calculation	Salvage and recovery	Member eligibility

Requirements

CLM-001: All domains MUST use shared claims infrastructure where the capabilities are genuinely common.

CLM-002: Each domain MUST own its claims adjudication logic.

CLM-003: Claim decisions MUST record the relevant policy or coverage references, decision rationale, evidence references, applicable rules, authority, and timestamps.

CLM-004: Claims reserves, approved amounts, payments, and recoveries MUST be represented as traceable financial movements.

CLM-005: Claims processing MUST safely handle duplicate submissions and retry scenarios.

PART III — SHARED ENTERPRISE SERVICES

13. Billing and Premium Receivables

13.1 Submodules

- Billing-account management.
- Premium schedule generation.
- Invoice and debit-note generation.
- Premium adjustments.
- Receivables management.
- Payment allocation.
- Arrears management.
- Credit notes.
- Refund requests.
- Receivables reconciliation.
- Collections reporting.

Requirements

BIL-001: The platform MUST generate billing obligations from authorised policy and premium transactions.

BIL-002: Billing records MUST reference the source policy, tenant, product, currency, and relevant business dates.

BIL-003: The system MUST support partial payments and payment allocation according to configured rules.

BIL-004: The platform MUST maintain an auditable history of billing adjustments and reversals.

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14. Payments and Settlements

14.1 Submodules

- Payment initiation.
- Payment-provider integration.
- Bank-transfer integration.
- Payment status management.
- Webhook and callback handling.
- Payment allocation.
- Refund processing.
- Reversal processing.
- Claim settlement.
- Provider payment.
- Commission payment.
- Payment reconciliation.

- Exception management.

Requirements

PAY-001: Payment operations MUST use idempotency controls to prevent duplicate financial effects.

PAY-002: Payment status changes MUST be traceable.

PAY-003: The system MUST support reconciliation between internal transactions and external bank or payment-provider records.

PAY-004: Payment credentials and sensitive payment data MUST be protected.

PAY-005: Failed or ambiguous payments MUST enter a controlled exception or reconciliation process rather than being silently treated as successful.

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15. Financial Subledger and Accounting Integration

The platform should include an insurance financial subledger while allowing integration with the insurer’s existing general ledger or ERP.

15.1 Submodules

- Transaction classification.
- Posting-rule configuration.
- Account mapping.
- Journal generation.
- Premium accounting.
- Claims accounting.
- Commission accounting.
- Reinsurance accounting interfaces.
- Reversals and adjustments.
- Accounting-period controls.
- Financial reconciliation.
- General ledger export.
- Posting exception management.

15.2 Requirements

FIN-001: Financial transactions MUST be traceable to their originating business transactions.

FIN-002: Posted entries MUST NOT be silently overwritten.

FIN-003: Corrections MUST use explicit reversal or adjustment transactions.

FIN-004: Posting rules and account mappings MUST be versioned and governed.

FIN-005: The subledger MUST support tenant, currency, accounting date, transaction date, source domain, and business reference as applicable.

FIN-006: Financial reconciliation MUST identify unmatched, duplicated, delayed, and inconsistent transactions.

FIN-007: Integration failures MUST be visible to authorised operational users.

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16. Commission and Distribution Management

16.1 Distribution channels

The platform should support:

- Direct sales.
- Brokers.
- Agents.
- Bancassurance.
- Employer distribution.
- Digital channels.
- Aggregators.
- Third-party distribution partners.

16.2 Commission submodules

- Intermediary registration.
- Appointment and licensing records.
- Distribution hierarchy.

- Product permissions.
- Commission-plan configuration.
- Commission calculation.
- Commission accrual.
- Commission adjustment.
- Commission statement.
- Commission settlement.
- Commission reconciliation.

Requirements

DST-001: Intermediaries MUST have controlled records and configurable product permissions.

DST-002: Commission rules MUST support effective dating and versioning.

DST-003: Commission calculations MUST retain the applicable plan version and source transaction.

DST-004: Commission payments MUST be subject to appropriate approval and financial controls.

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17. Reinsurance Framework

The platform should support reinsurance as a shared enterprise capability, with domain-specific exposure and claim data supplied by Life, General, and Health.

17.1 Submodules

- Reinsurer records.
- Treaty administration.
- Facultative arrangement records.
- Treaty effective dates.
- Cession rules.
- Exposure allocation.
- Premium cessions.
- Claims recoveries.
- Reinsurance statements.
- Bordereaux.
- Reconciliation.
- Reinsurance reporting.

Requirements

REI-001: Reinsurance records MUST retain links to the underlying policies, risks, claims, and financial transactions.

REI-002: Cession calculations MUST be reproducible using the applicable arrangement and configuration versions.

REI-003: Reinsurance recoverables and settlements MUST be traceable and reconcilable.

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18. Document and Content Management

18.1 Submodules

- Document generation.
- Template management.
- Template versioning.
- Document storage.
- Metadata management.
- Document retrieval.
- Access control.
- Evidence management.
- Document retention.
- Legal hold.
- Document delivery.
- Document audit.

18.2 Document categories

- Proposals.
- Policies.
- Schedules.
- Certificates.
- Endorsements.
- Premium notices.
- Claim evidence.
- Medical documents.
- Assessment reports.
- Settlement letters.
- Payment remittance advice.

- Regulatory documents.

Requirements

DOC-001: Documents MUST be tenant-scoped and access-controlled.

DOC-002: Generated documents MUST retain references to the applicable template and product versions.

DOC-003: Access to sensitive documents MUST be auditable.

DOC-004: Medical and other highly sensitive documents MUST receive additional access controls appropriate to their classification.

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19. Workflow and Case Management

The platform should provide shared workflow infrastructure that supports domain-specific processes.

19.1 Submodules

- Workflow definition.
- Workflow versioning.
- State and transition management.
- Task assignment.
- Work queues.
- Maker-checker approvals.
- Delegated authority.
- SLA timers.
- Escalation.
- Exception management.
- Workflow audit.
- Case history.

Requirements

WFL-001: Workflow definitions MUST be versioned.

WFL-002: A workflow transition MUST invoke the owning domain’s business validation.

WFL-003: Workflows MUST support assignment, reassignment, escalation, and controlled delegation.

WFL-004: Sensitive transitions MUST enforce approval authority.

WFL-005: Long-running workflows MUST support recovery after service restarts or temporary integration failures.

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20. Integration and API Management

The platform must provide a governed integration layer for internal modules and external systems.

20.1 Integration submodules

- API gateway.
- API catalogue.
- API authentication.
- Rate limiting.
- Partner registration.
- Integration credentials.
- Webhook management.
- Event messaging.
- Event schema governance.
- Retry and dead-letter handling.
- Integration monitoring.
- Error handling.
- Reconciliation and replay.

20.2 Potential integrations

Integration category	Examples
Identity	Enterprise identity provider, workforce SSO

Payments	Banks, payment gateways, mobile money providers
Finance	General ledger, ERP, accounting platforms
Distribution	Broker systems, agent systems, bancassurance platforms
Healthcare	Provider systems, claims clearinghouses, coding systems
Reinsurance	Reinsurer platforms, treaty administration systems
Documents	Electronic signature and document-delivery services
Communications	Email, SMS, approved messaging providers
Analytics	Data warehouse, business intelligence tools
Regulatory	Approved regulatory submission and reporting interfaces

20.3 API requirements

INT-001: APIs MUST be documented and versioned.

INT-002: APIs MUST enforce authentication, authorisation, tenant scope, and input validation.

INT-003: Non-repeatable business operations MUST support idempotency where appropriate.

INT-004: Events MUST have stable identifiers and versioned schemas.

INT-005: Event consumers MUST tolerate duplicate delivery.

INT-006: Failed integrations MUST support controlled retry, monitoring, and recovery.

INT-007: Event payloads MUST minimise sensitive data.

PART IV — DATA MODEL AND DOMAIN OWNERSHIP

21. Core Data Model

The platform should use a shared Party model and domain-owned insurance aggregates.

21.1 Shared conceptual entities

Entity	Responsibility
Tenant	Tenant identity, configuration, isolation boundary
Party	Individual or organisation
PartyRole	A party's role in an insurance or business relationship
ProductDefinition	Logical product identity
ProductVersion	Immutable approved product configuration
RatingVersion	Version of rating configuration or calculation implementation

RuleSetVersion	Version of underwriting or decision rules
WorkflowVersion	Version of a business process
TemplateVersion	Version of a document template
DocumentReference	Controlled reference to stored content
ConsentRecord	Consent and preference records
BillingAccount	Billing relationship
Invoice	Billing obligation
Payment	Payment instruction and status
PaymentAllocation	Allocation of a payment to an obligation
Commission	Commission entitlement or transaction
SubledgerEntry	Detailed financial posting
AuditEvent	Auditable record of an operation

21.2 Domain-owned aggregates

The domains should own their primary transactional records.

Life domain

- LifeProposal.
- LifeUnderwritingCase.
- LifePolicy.
- LifeBenefit.
- LifeBeneficiary.
- LifePremiumSchedule.
- LifeCashValue.
- LifePolicyLoan.
- LifeClaim.

General domain

- GeneralQuote.
- GeneralProposal.
- GeneralRiskObject.
- GeneralPolicy.
- GeneralCoverage.
- GeneralEndorsement.
- GeneralClaim.
- LossAssessment.
- RecoveryCase.

Health domain

- HealthScheme.
- HealthPlan.
- HealthMembership.
- HealthBenefit.
- ProviderContract.
- ProviderTariff.
- Preauthorisation.
- MedicalClaim.
- UtilisationCase.

The exact aggregate boundaries should be validated during detailed domain modelling.

21.3 Data ownership rules

1. Each domain owns its authoritative transactional data.
2. Other domains should not write directly into another domain's database.
3. Cross-domain interactions should use defined APIs or events.
4. Reporting systems may maintain derived projections but must not become the source of transactional truth.
5. Historical transactions must retain references to the product and configuration versions that governed them.
6. Data access must respect tenant and sensitivity boundaries.

PART V — API AND EVENT ARCHITECTURE

22. API Design

REST/JSON is a reasonable default for synchronous business APIs, subject to the final technology and integration strategy.

22.1 Illustrative API groups

API group	Example operations
Tenant	Create tenant, configure tenant, suspend tenant
Party	Create party, update party, search party, manage relationships
Product	Create product version, validate, simulate, approve, publish
Quotation	Create quote, calculate premium, refer to underwriting

Life Policy	Issue policy, change beneficiary, reinstate, surrender
General Policy	Issue policy, endorse, renew, cancel
Health Membership	Enrol member, add dependant, verify eligibility
Preauthorisation	Submit request, assess, approve, reject
Claims	Register, assess, approve, settle, close
Billing	Generate invoice, retrieve receivables, allocate payment
Payments	Initiate payment, process callback, reconcile
Finance	Post subledger entry, reverse transaction, reconcile
Reporting	Run report, retrieve result, export authorised data

These are logical API groups, not final endpoint contracts.

22.2 API requirements

Each API contract should define:

- Authentication and authorisation requirements.
- Tenant scope.
- Request and response schemas.
- Validation rules.
- Error codes.

- Idempotency behaviour.
- Rate limits.
- Pagination.
- Versioning policy.
- Audit requirements.
- Data sensitivity classification.

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23. Event-Driven Integration

Events should support decoupled operations and asynchronous processes.

23.1 Illustrative domain events

Product events

- ProductVersionValidated.
- ProductVersionApproved.
- ProductVersionPublished.

Life events

- LifeProposalSubmitted.
- LifePolicyIssued.
- LifePolicyChanged.
- LifePolicyLapsed.
- LifeClaimRegistered.
- LifeClaimSettled.

General events

- GeneralQuoteAccepted.
- GeneralPolicyIssued.
- GeneralPolicyEndorsed.
- GeneralPolicyRenewed.
- GeneralClaimRegistered.
- GeneralClaimSettled.

Health events

- HealthMemberEnrolled.

- HealthMemberStatusChanged.
- EligibilityDecisionRecorded.
- PreauthorisationApproved.
- MedicalClaimSubmitted.
- MedicalClaimSettled.

Financial events

- InvoiceIssued.
- PaymentReceived.
- PaymentAllocated.
- PaymentReversed.
- CommissionAccrued.
- SubledgerEntryPosted.
- ReconciliationExceptionRaised.

23.2 Event requirements

Events MUST include appropriate metadata, such as:

- Event identifier.
- Tenant identifier.
- Event type and version.
- Occurrence timestamp.
- Aggregate reference.
- Correlation identifier.
- Causation identifier where applicable.

Consumers must tolerate duplicate delivery.

For important transactions, use a transactional outbox or equivalent reliable publication mechanism so a committed business transaction is not silently lost between the database and the event broker.

PART VI — SECURITY, PRIVACY, AND COMPLIANCE

24. Security Requirements

The platform must implement:

- Least-privilege access.
- Deny-by-default authorisation.
- Secure credential management.
- Encryption in transit and at rest.
- Controlled key management.
- API authentication and rate limiting.
- Privileged access controls.
- Secure development practices.
- Dependency and vulnerability scanning.
- Security monitoring.
- Incident response.
- Backup protection.
- Tenant-isolation testing.

Critical security findings must be addressed before production acceptance, subject only to a formally approved exception and remediation plan.

24.1 Sensitive data

The platform should classify information into categories such as:

- Public.
- Internal.
- Confidential.
- Personal.
- Highly sensitive.

Medical records and health claims evidence should receive heightened access restrictions.

Sensitive data must be minimised in logs, event messages, support tickets, and analytics datasets.

24.2 Regulatory localisation

The platform must support a jurisdiction-specific compliance register.

Before launching in each market, the implementation team must validate applicable requirements concerning:

- Insurance licensing and supervision.
- Policy wording and disclosures.
- Claims handling.

- Data protection and privacy.
- Data residency.
- Medical information.
- Taxation.
- Electronic signatures.
- Accounting and financial reporting.
- Outsourcing and cloud hosting.
- Regulatory submissions.
- Record retention.

This specification establishes design requirements; it does not itself certify compliance with any particular jurisdiction.

PART VII — NON-FUNCTIONAL REQUIREMENTS

25. Availability and Performance

The following are proposed starting targets, not confirmed contractual commitments.

ID	Attribute	Proposed target
NFR-001	Availability	99.9% monthly availability for core production services
NFR-002	Standard API latency	p95 ≤ 2 seconds for agreed standard operations under benchmark load
NFR-003	Standard rating latency	p95 ≤ 3 seconds for agreed standard rating scenarios

NFR-004	Scalability	Horizontal scaling for stateless and high-volume components
NFR-005	Recovery Point Objective	Proposed RPO ≤ 15 minutes for core transactional data
NFR-006	Recovery Time Objective	Proposed RTO ≤ 4 hours for a regional disaster scenario
NFR-007	Auditability	End-to-end traceability for critical policy, claim, configuration, and financial operations
NFR-008	Maintainability	Documented module interfaces, automated tests, and clear ownership
NFR-009	Interoperability	Versioned APIs and governed integration contracts
NFR-010	Accessibility	Target WCAG 2.2 AA for web interfaces where applicable
NFR-011	Portability	Automated deployment with limited unnecessary dependence on cloud-specific services
NFR-012	Observability	Centralised metrics, logs, traces, dashboards, and alerts

NFR-013	Data integrity	Domain invariants enforced within authoritative transactional boundaries
NFR-014	Backup	Encrypted, monitored, access-controlled, and restore-tested backups
NFR-015	Capacity	Load tests covering peak quotes, issuance, claims, provider submissions, and month-end processing

These targets must be validated against representative workloads before they are adopted as customer-facing service commitments.

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26. Cloud Deployment and Operations

26.1 Environment strategy

The platform should maintain separate:

- Development environments.
- Test environments.
- Staging or pre-production environments.
- Production environments.

26.2 Deployment requirements

The platform should use:

- Infrastructure as code.
- Automated build and deployment pipelines.
- Controlled database migrations.
- Versioned application releases.
- Secrets management.
- Automated testing gates.
- Dependency and container scanning.

- Release and rollback procedures.
- Post-deployment verification.

Production configuration changes must be controlled and auditable.

26.3 Observability

The operational platform should monitor:

- Availability.
- API latency.
- Error rates.
- Database saturation.
- Queue lag.
- Failed events.
- Payment exceptions.
- Integration failures.
- Product publication failures.
- Tenant-isolation violations.
- Backup status.
- Resource consumption.

Operational dashboards must not expose unauthorised tenant data.

PART VIII — TESTING AND ACCEPTANCE

27. Testing Strategy

The platform must support multiple testing layers.

27.1 Unit testing

Test:

- Domain rules.
- Rating calculations.
- Underwriting decisions.
- Policy state transitions.
- Benefit calculations.

- Claim adjudication.
- Financial posting rules.

27.2 Integration testing

Test:

- Database interactions.
- Document storage.
- Event messaging.
- Identity providers.
- Payment providers.
- Accounting integrations.
- External healthcare and distribution systems.

27.3 End-to-end testing

The minimum end-to-end scenarios should include:

1. Product configuration to publication.
2. Quote to policy issuance.
3. Policy issuance to premium billing.
4. Payment receipt to allocation and accounting.
5. Claim registration to settlement.
6. Health member enrolment to eligibility verification.
7. Health preauthorisation to medical claim processing.
8. Financial transaction to subledger and reconciliation.

27.4 Security testing

The test suite must include:

- Cross-tenant access attempts.
- Privilege escalation.
- Unauthorised document retrieval.
- Cross-tenant cache and search tests.
- Event-consumer tenant validation.
- API authorisation bypass attempts.
- Sensitive-data leakage.
- Credential and secrets exposure.

27.5 Production acceptance criteria

The platform must not be accepted for production until:

- Mandatory tenant-isolation tests pass.
- Critical policy and financial invariants are tested.
- Product configuration publication is governed and auditable.
- Approved financial scenarios reconcile.
- Integration failures have defined recovery procedures.
- Monitoring and alerting are operational.
- Backup restoration has been demonstrated.
- Disaster-recovery targets have been tested.
- Relevant regulatory and compliance reviews have been completed.
- The agreed production workload meets the approved performance and availability targets.

PART IX — DELIVERY ROADMAP

28. Recommended Implementation Phases

The platform should be delivered incrementally, with each phase producing a demonstrable business outcome.

Phase 0: Discovery and architecture validation

Deliverables

- Market and jurisdiction analysis.
- Product taxonomy.
- Initial domain model.
- Tenant-isolation architecture.
- Security and compliance register.
- Target workloads.
- Architecture decision records.
- MVP scope.
- Prioritised requirements backlog.

Exit criteria

The product owner, insurance specialists, architecture team, and delivery leadership approve the scope, architecture, and acceptance criteria for the first production release.

Phase 1: Platform foundation

Modules

- Tenant Management.
- Identity and Access.
- Party Management.
- Product Catalogue foundation.
- Configuration Governance.
- Audit.
- API Gateway.
- Integration foundation.
- Observability.
- Document Management foundation.
- Deployment automation.

Exit criteria

A tenant can be provisioned securely, authorised users can access permitted capabilities, and automated tests demonstrate tenant isolation.

Phase 2: Shared insurance services

Modules

- Product Factory.
- Rating framework.
- Rules engine.
- Workflow.
- Billing.
- Payments.
- Commission framework.
- Financial subledger.
- Notifications.
- Shared claims infrastructure.

Exit criteria

A configured product can progress through an agreed quotation, policy issuance, billing, payment, and accounting workflow.

Phase 3: First insurance domain

Implement one domain end-to-end.

The selection should be based on commercial commitments, target-market demand, domain expertise, and the ability to deliver a complete production-grade product.

Exit criteria

The selected domain supports a complete agreed business journey, including policy administration, claims, financial processing, reporting, and operational controls.

Phase 4: Remaining insurance domains

Implement the other two domains, reusing shared platform services while preserving domain-specific business logic.

Exit criteria

Each domain meets its own functional acceptance suite and remains independently evolvable.

Phase 5: Enterprise capabilities

Introduce prioritised advanced capabilities, including:

- Reinsurance.
- Advanced actuarial integrations.
- Regulatory reporting.
- Fraud analytics.
- Enterprise data and analytics.
- Advanced reconciliation.
- Expanded distribution integrations.

Phase 6: Scale and optimisation

Focus on:

- Performance optimisation.
- Tenant service tiers.
- Operational automation.
- Advanced disaster recovery.
- Security maturity.
- Product configuration governance.
- Capacity planning.

PART X — KEY RISKS AND MITIGATIONS

29. Risk Register

Risk	Potential impact	Mitigation
Over-generalised policy model	Complex rules and poor domain evolution	Separate Life, General, and Health policy aggregates
Tenant data leakage	Severe legal and reputational impact	Defence in depth, automated isolation tests, controlled support access
Uncontrolled configuration	Incorrect pricing or coverage	Validation, simulation, approvals, versioning, effective dates
Premature microservices	Operational complexity and slower delivery	Begin with modular boundaries and split services where justified
Unclear regulatory scope	Delayed launch or non-compliance	Jurisdiction-specific compliance analysis before production
Financial inconsistencies	Incorrect balances and settlements	Idempotency, reconciliation, controlled reversals, traceable postings
External integration failures	Delays and operational backlogs	Retries, timeouts, dead-letter handling, monitoring, runbooks

Scope expansion	Long delivery cycles	Deliver one complete vertical slice before broad expansion
Unvalidated performance targets	Cost overruns or SLA failures	Benchmark representative workloads before committing to SLAs
Insufficient configuration governance	Unauthorised product changes	Maker-checker, immutable versions, simulation, audit

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PART XI — OPEN DESIGN DECISIONS

30. Decisions Required Before Detailed Technical Design

The core product architecture is established, but the following decisions require stakeholder input.

Decision	Question to resolve
First production domain	Which of Life, General, or Health should be delivered first?
Initial markets	Which countries and regulators are in scope for the first release?

Cloud providers	Which cloud providers and regions are approved?
Tenant service tiers	Which customers require dedicated databases or environments?
Service levels	What availability, latency, throughput, RPO, and RTO are contractual requirements?
Accounting	Which accounting standards, chart-of-account structures, and tax rules apply?
Currency	Which currencies and multi-currency accounting scenarios are required?
Health standards	Which medical coding systems and provider claim formats are required?
Actuarial calculations	Which calculations must be native and which may use external engines?
Integrations	Which payment, identity, accounting, distribution, and regulatory systems must be integrated first?
Data retention	What retention, deletion, and legal-hold requirements apply in each jurisdiction?
Workload	What are the target tenant count, policies in force, quote volume, claims volume, and concurrency?

Technology stack	Which languages, frameworks, databases, message brokers, and frontend technologies are preferred or mandated?

These are not blockers to the conceptual architecture, but they are necessary for finalising implementation-level designs and delivery estimates.

PART XII — REQUIREMENTS TRACEABILITY MATRIX

31. Capability-to-Acceptance Mapping

Capability	Key requirements	Minimum acceptance evidence
Tenant isolation	TEN-001 to TEN-010	Automated tests prove that unauthorised cross-tenant access is blocked
Identity and access	IAM-001 to IAM-007	Role tests, tenant-scoped access tests, and privileged-operation audit
Product configuration	PRD-001 to PRD-010	Product validates, is approved and published, and remains reproducible
Rating and underwriting	RAT requirements	Approved test cases reproduce expected pricing and decision outcomes

Life insurance	LIF requirements	Complete Life quote-to-policy-to-claim scenario
General insurance	GEN requirements	Complete General quote-to-policy-to-claim scenario
Health insurance	HLT requirements	Complete member-to-eligibility-to-claim scenario
Billing and payments	BIL and PAY requirements	Premium billing, payment allocation, reversal, and reconciliation tests
Financial accounting	FIN requirements	Subledger posting and external accounting reconciliation
Integration	INT requirements	API contract, retry, duplicate-event, and recovery tests
Security and compliance	Security and compliance requirements	Security assessment, audit review, and jurisdiction-specific sign-off
Operations	NFR and deployment requirements	Load, backup, restore, monitoring, and disaster-recovery evidence

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PART XIII — ARCHITECTURE DECISION RECORDS

32. Recommended Architecture Decision Register

The following records should be created and maintained as the architecture evolves.

ADR	Decision
ADR-001	Tenant isolation strategy and deployment tiers
ADR-002	Initial application topology and service decomposition
ADR-003	Domain boundaries and authoritative data ownership
ADR-004	Product configuration schemas and rules engine
ADR-005	API versioning and event schema governance
ADR-006	Identity provider and tenant-context propagation
ADR-007	Financial posting, reversal, and reconciliation model
ADR-008	Cloud region, backup, RPO, RTO, and disaster-recovery design
ADR-009	Data classification, medical-data handling, and retention

ADR-010	First production domain and MVP acceptance criteria

Each ADR should record the context, decision, alternatives considered, rationale, consequences, risks, and approval.

PART XIV — FINAL ARCHITECTURAL RECOMMENDATIONS

33. Five decisions that should guide engineering

33.1 Treat tenant isolation as a platform capability

Do not implement tenant isolation independently in each insurance module. Establish trusted tenant context, authorisation policies, data-access enforcement, and isolation testing in the platform foundation.

33.2 Keep the insurance domains separate

Life, General, and Health must have their own policy aggregates, state machines, and claims adjudication logic.

Share technical and business capabilities where they are genuinely common, but avoid creating a generic policy model that becomes increasingly complex as each domain adds exceptions.

33.3 Build the Product Factory early

The Product Factory is central to the platform's commercial value.

Product definitions, rates, benefits, rules, workflows, and document templates should be versioned and governed from the beginning. Retrofitting configuration governance after products have been hard-coded will create unnecessary complexity.

33.4 Establish financial traceability from the beginning

Every premium, commission, claim payment, refund, recovery, and financial adjustment must be traceable to its originating business transaction.

Use explicit reversals and adjustments rather than silently modifying posted financial records.

33.5 Deliver one complete insurance journey before expanding broadly

Do not build three large collections of partially implemented modules in parallel without proving the architecture.

Deliver a complete vertical slice covering product configuration, quotation, underwriting, policy issuance, billing, payment, claims, settlement, accounting, and reporting for the first selected domain. Then expand into the other domains using validated shared capabilities.

Conclusion

The proposed Unified Insurance Platform is best understood as a **shared insurance technology platform with three independent insurance domains**, rather than three separate systems connected through a common database.

Its architectural foundation consists of:

1. Strict tenant isolation.
2. Cloud-ready deployment.
3. Separate Life, General, and Health policy models.
4. Configuration-driven product management.
5. Shared platform and enterprise services.
6. Domain-owned claims adjudication.
7. Traceable billing, payments, and financial accounting.
8. Governed APIs and event integration.
9. Security, audit, and compliance by design.
10. Incremental implementation with measurable acceptance criteria.

This specification establishes the functional baseline for detailed solution design. The next engineering artefacts should be the **domain model and entity relationship diagrams, module-level API contracts, detailed workflow and state-transition specifications, database architecture, and prioritised implementation backlog**.

One important qualification remains: the requirements and proposed performance targets must be validated against the initial target markets, regulatory obligations, workload assumptions, and customer service commitments before being treated as final contractual specifications.